

A Counterexample to Conjecture 00000007976: the Radius-1 Covering Density Is Not $1 + 2^{-n}$

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Abstract

The third claim of Conjecture 00000007976 states that the optimal density of binary covering codes of radius 1 is exactly $\mu(n, 1) = 1 + 2^{-n}$. For every odd n this is impossible for any code: the density of a code $C \subseteq \mathbb{F}_2^n$ is $|C|(n+1)/2^n$, and $|C|(n+1)$ is even, while $2^n(1 + 2^{-n}) = 2^n + 1$ is odd. In fact $\mu(3, 1) = \mu(7, 1) = 1$, attained by the repetition and Hamming codes. Both results are checked in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000007976 reads:

Definition: The covering density constant of covering codes: the density $\mu(n, r)$ of radius- r covering codes of binary space (the ratio of the counted ball volume to the space volume). Conjecture: The infimum μ_n is $1 + \Theta(n^{-1}2^{-n})$ (existence of asymptotically perfect coverings); and the limit structure of optimal covering codes (n to infinity, r fixed) is a generalization of lattice coverings whose normalized second moment of the covering radius attains the Zador constant; and the exact value of $\mu(n, 1)$ is $1 + 2^{-n}$ (the unique realization of the radius-1 Rosenbloom type).

The Chinese version in `conjectures/00000007976.md` states the same three claims.

2 Reading of the statement

The conjecture is a conjunction of three claims. We refute the third one, so the conjecture is false whatever the first two claims mean.

Definition 1. A code $C \subseteq \mathbb{F}_2^n$ covers with radius r if every word of $\mathbb{F}_2^n$ is within Hamming distance r of some codeword. Its density is $|C|V(n, r)/2^n$, where $V(n, r) = \sum_{i \leq r} \binom{n}{i}$ is the volume of a Hamming ball of radius r . Let $K(n, r)$ be the least size of a covering code of radius r , and $\mu(n, r) = K(n, r)V(n, r)/2^n$ the optimal density.

The third claim states $\mu(n, 1) = 1 + 2^{-n}$, which we read as holding for every $n \geq 1$. We also refute the weaker reading “for all sufficiently large n ”. If $\mu(n, 1)$ is read instead as the density of some radius-1 code, the claim fails as well, because Theorem 2 applies to every code.

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3 Results

Theorem 2. *Let n be odd. Then no code $C \subseteq \mathbb{F}_2^n$ has density $|C|(n+1)/2^n$ equal to $1 + 2^{-n}$. In particular $\mu(n, 1) \neq 1 + 2^{-n}$ for every odd n .*

Theorem 3. $\mu(3, 1) = 1$ and $\mu(7, 1) = 1$.

Theorem 2 refutes the third claim at $n = 1$, and for arbitrarily large n ; so the claim fails for every $n \geq 1$ as stated and for all sufficiently large n . Theorem 3 shows the true values in two cases.

4 Proofs

Proof of Theorem 2. Since $V(n, 1) = n + 1$, the equation $|C|(n+1)/2^n = 1 + 2^{-n}$ is equivalent to $|C|(n+1) = 2^n + 1$. For odd n the left side is even, because $n+1$ is even, while the right side is odd, because $n \geq 1$. The same argument applies to $K(n, 1)$. $\square$

Lemma 4 (sphere-covering bound). *If C covers $\mathbb{F}_2^n$ with radius 1, then $|C|(n+1) \geq 2^n$.*

Proof. The ball of radius 1 around c consists of c and the n words $c + e_i$, so it has at most $n+1$ elements. These balls cover the 2^n words of $\mathbb{F}_2^n$. $\square$

Proof of Theorem 3. If a radius-1 covering code C has $|C|(n+1) = 2^n$, then $K(n, 1) = |C|$ by Lemma 4, so $\mu(n, 1) = 1$. For $n = 3$ take the repetition code $\{000, 111\}$: every word has at least two equal coordinates and is therefore within distance 1 of one of the two codewords; $2 \cdot 4 = 8$. For $n = 7$ take the Hamming code, the solutions of $x_1 + x_3 + x_5 + x_7 = x_2 + x_3 + x_6 + x_7 = x_4 + x_5 + x_6 + x_7 = 0$ over $\mathbb{F}_2$. It has 16 codewords and covers with radius 1, since flipping the coordinate whose binary index equals the syndrome corrects any word; $16 \cdot 8 = 128$. $\square$

5 Formalization

The Lean 4 project in `lean/` (file `lean/Submission/Basic.lean`) formalizes the definitions above and proves both theorems against Mathlib.

- Words are `Fin n → ZMod 2`, and distance is Mathlib's `hammingDist`. `Covers`, `ballVolume`, `density`, `coveringNumber` (as `sInf`) and `mu` follow the definition above.
- `Claim3` is the third claim for every $n \geq 1$; `Claim3Eventually` is the claim for all large n .
- `parity_contra`, `density_ne` and `mu_ne` give Theorem 2; `not_claim3` and `not_claim3Eventually` refute both readings.
- `sphere_covering` is Lemma 4, and `mu_eq_one_of_perfect` turns a perfect covering into $\mu(n, 1) = 1$. `rep3` and `hamming7` are the two codes, whose covering property and size are checked by `decide`; this gives `mu_three` and `mu_seven`.
- `conjecture_00000007976_false` states, for arbitrary propositions P, Q standing for the first two claims, that $\neg(P \wedge Q \wedge \text{Claim3})$ and $\neg(P \wedge Q \wedge \text{Claim3Eventually})$.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms: `#print axioms` reports exactly `propext`, `Classical.choice` and `Quot.sound` for `conjecture_00000007976_false` and `mu_seven`.

6 Remarks

Remark 5. More generally the Hamming code of length $n = 2^m - 1$ is a perfect radius-1 code, so $\mu(2^m - 1, 1) = 1$ for every $m \geq 2$; the case $m = 1$ is the trivial code $\{0\} \subseteq \mathbb{F}_2$. We formalize $m = 2, 3$ only. These perfect codes also bear on the first claim, since $\mu(n, 1) - 1 = 0$ for infinitely many n , but we do not use this because the first claim does not fix what μ_n is the infimum over.